CUBIC LOWER BOUNDS FOR CONJUGACY SEPARATION IN NILPOTENT GROUPS

Abstract. This paper proposes a construction of a torsion-free finitely generated nilpotent group of Hirsch length $8d$ for every integer $d \geq 2$, with a claimed lower bound $n^{3d^2(d+1)}$ for its conjugacy separability growth. This would give a negative answer to Question 1 of Deré and Vandeputte on the existence of a universal quadratic bound, in Hirsch length, for the degree of this growth function. Together with their upper bound, it would show that cubic order in Hirsch length is necessary and sufficient for a universal degree bound, up to absolute constants. The argument uses integral semidirect products, an elementary word compression estimate, and characters of finite abelian normal subgroups. It addresses arbitrary finite quotients separating the selected pairs and gives explicit finite separating quotients.

1. Introduction

Let G be a finitely generated group, and let S be a finite generating set. We write $|g|_S$ for word length with respect to $S \cup S^{-1}$. For nonconjugate elements $a, b \in G$, their conjugacy separation depth is

$$D_G(a,b) = \min\{|Q| : \ \varphi : G \twoheadrightarrow Q, \quad Q \text{ finite}, \quad \varphi(a) \not\sim_Q \varphi(b)\},$$

where $\sim_Q$ denotes conjugacy in Q. We allow the value $+\infty$ if no such quotient exists. The conjugacy separability growth function is

$$\mathrm{Conj}_{G,S}(n) = \max\{D_G(a,b) : \ |a|_S, |b|_S \leq n, \quad a \not\sim_G b\},$$

with value 1 when the set is empty. Finitely generated nilpotent groups are conjugacy separable, so this function is finite for the groups under consideration; see [1, Introduction].

For nondecreasing functions on the positive integers, write $f \preceq g$ if there is an integer $C \geq 1$ such that $f(n) \leq Cg(Cn)$ for every $n \geq 1$, and write $f \simeq g$ if both comparisons hold. We use $f \succeq g$ to mean $g \preceq f$. Changing the finite generating set does not change the resulting growth class, since the corresponding word lengths are bounded by constant multiples of one another. We therefore omit S in statements made up to this comparison. All multiplicative constants may depend on the fixed group and generating set.

Deré and Vandeputte prove an upper bound for this function in terms of the nilpotency class and Hirsch length [1, Theorem A]. They ask whether there is an absolute constant $C > 0$ such that

$$\mathrm{Conj}_G(n) \preceq n^{Ch(G)^2} \tag{1.1}$$

for every torsion-free finitely generated nilpotent group G [1, Question 1, Section 3.2]. Here $h(G)$ denotes the Hirsch length. The proposed main result is the following.

Theorem 1.1. *For every integer $d \geq 2$, there is a torsion-free finitely generated nilpotent group $\mathcal{G}_d$ of class $4d$ and Hirsch length $8d$ such that*

$$\mathrm{Conj}_{\mathcal{G}_d}(n) \succeq n^{3d^2(d+1)}.$$

More precisely, for every finite generating set S of $\mathcal{G}_d$, there are constants $c_{d,S} > 0$ and $n_{d,S} \geq 1$ such that

$$\mathrm{Conj}_{\mathcal{G}_d,S}(n) \geq c_{d,S} n^{3d^2(d+1)} \qquad \text{for every integer } n \geq n_{d,S}.$$

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Consequently, no absolute constant C satisfies (1.1).

The ratio of the exponent in Theorem 1.1 to the square of the Hirsch length is

$$\frac{3d^2(d+1)}{(8d)^2} = \frac{3(d+1)}{64},$$

which tends to infinity. The lower bound is obtained by fixing a single group $\mathcal{G}_d$ and a single prime $p > 4d$, and then varying an integer parameter $s \geq 1$ in a sequence of pairs of elements.

The claimed lower bound would also determine the order of the optimal universal exponent bound as a function of Hirsch length.

Corollary 1.2. *Every torsion-free finitely generated nilpotent group G satisfies*

$$\mathrm{Conj}_G(n) \preceq n^{h(G)^3}.$$

There is an absolute constant $c_0 > 0$ and a sequence of such groups G with unbounded Hirsch length for which

$$\mathrm{Conj}_G(n) \succeq n^{c_0 h(G)^3}.$$

In particular, no function $E : \mathbb{N}_0 \to [0, \infty)$ with $E(h) = o(h^3)$ can satisfy $\mathrm{Conj}_G(n) \preceq n^{E(h(G))}$ for all torsion-free finitely generated nilpotent groups G.

Proof. Suppose first that G is nonabelian. Put $h = h(G)$, let k be its nilpotency class, and put $z = h(Z(G))$ and $r_0 = h([G, G])$. By [1, Theorem A],

$$\mathrm{Conj}_G(n) \preceq n^{(k-1)(h-z+1)r_0}.$$

The inequalities $k - 1 \leq h$, $z \geq 1$, and $r_0 \leq h$ bound this exponent by h^3. If G is nontrivial abelian, its growth is equivalent to $\log n$ by [1, Proposition 2.1.9]; the trivial group has constant growth. This proves the upper bound. For the lower bound, take $G = \mathcal{G}_d$ and $c_0 = 3/512$, since

$$\frac{3d^2(d+1)}{(8d)^3} = \frac{3(d+1)}{512d} \geq \frac{3}{512}.$$

For any proposed function $E = o(h^3)$, choose one sufficiently large d so that $E(8d) < c_0(8d)^3$. The two polynomial bounds in this fixed group are then incompatible as $n \to \infty$. □

The groups in the proof have the form $G(m) = \mathbb{Z}^m \rtimes H$, where H is itself an abelian-by-cyclic nilpotent group. Two actions on the normal subgroup $\mathbb{Z}^m$ play different roles. One produces a bidiagonal conjugacy equation; the other forces sufficiently many matrix units to act on the abelian subgroup in every finite quotient. A character detecting the chosen conjugacy obstruction then gives many independent characters of large order. Section 2 supplies the word estimates, and Sections 3–7 give the construction and quotient argument.

Throughout, group commutators are $[x, y] = xyx^{-1}y^{-1}$, vectors are columns, and exponentials and logarithms of nilpotent or unipotent matrices are finite polynomial expressions.

Formalization and verification. The companion repository contains the Lean sources, reproduction instructions, and verification records. At revision `b017c9e` (6 October 2026), the declaration `cubicFamily_main` proves the group construction, exact nilpotency class, an explicit series with $8d$ infinite cyclic quotients, and the polynomial lower bound in every finite word metric. Its witnesses are nonconjugate pairs with finite separating depth. The declaration `no_universal_quadratic_exponent` formalizes the negative answer to the proposed quadratic bound. The formal proof uses the subsequence $s = wk$ of the pairs below, which suffices for the same lower exponent at all sufficiently large radii. The complete coverage and limitations are recorded in the verification scope; in particular, the general conjugacy-separability theorem and the cited upper bound in Corollary 1.2 are not formalized there. Build, kernel replay, and axiom

checks passed for this revision. These checks concern the formal statements; their correspondence with the manuscript is a separate question.

2. An elementary word compression estimate

We first prove the word estimate needed in the construction. All word length comparisons in this section are upper bounds.

Lemma 2.1. *Let $l \geq 1$ be an integer and let $F_l = \mathbb{Z}^l \rtimes_A \mathbb{Z}$, where $A = I + N$ and*

$$Ne_i = e_{i+1} \quad (1 \leq i < l), \qquad Ne_l = 0.$$

Use the generating set $e_1, \dots, e_l, t$, where t generates the cyclic factor. For every $1 \leq j \leq l$ and every $a \in \mathbb{Z}$,

$$|ae_j|_{F_l} \leq C_l(1 + |a|^{1/j}).$$

Proof. We use descending induction on j. It suffices to consider $a > 0$: inversion handles negative a, and $a = 0$ is immediate. Put $B = \lfloor a^{1/j} \rfloor + 1 \geq 2$. Since $a < B^j$, write

$$a = \sum_{u=0}^{j-1} a_u B^u, \qquad 0 \leq a_u < B.$$

The commutator identity

$$[t^v, x] = (A^v - I)x \qquad (x \in \mathbb{Z}^l)$$

shows that iterated commutators with u copies of t^B and $j - 1 - u$ copies of t, applied to $a_u e_1$, represent

$$x_u = a_u (A^B - I)^u N^{j-1-u} e_1.$$

The operators commute, so their order in the iteration is immaterial. Each such word has length $O_l(B)$: its depth is at most $l - 1$, and every input word has length at most B. When $j = 1$, this simply means the word representing $a_0 e_1$.

The first nonzero possible coordinate of x_u is its e_j coordinate, which is $a_u B^u$. To control the higher coordinates, expand

$$A^B - I = \sum_{v=1}^{l-1} \binom{B}{v} N^v.$$

For a contribution to the e_i coordinate with $i > j$, the indices from the u factors sum to $i - j + u$. Its coefficient is therefore bounded by $C_l B^{1+i-j+u}$, including the digit a_u. As $u \leq j - 1$, this is at most $C_l B^i$. Consequently the sum of the j vectors is

$$\sum_{u=0}^{j-1} x_u = ae_j + \sum_{i>j} b_i e_i, \qquad |b_i| \leq C_l B^i,$$

and has a representative of length $O_l(B)$.

For $j = l$ there are no error coordinates, which starts the induction. For smaller j, the induction hypothesis represents every $b_i e_i$ with $i > j$ by a word of length $O_l(B)$. Cancel these finitely many vectors. They all lie in the abelian subgroup $\mathbb{Z}^l$, so the cancellations introduce no further errors. This gives a word of length $O_l(B)$ for ae_j, as required. □

Lemma 2.2. *Let $l \geq 1$ be an integer and let $K = \mathbb{Z}^l \rtimes_A \mathbb{Z}$, where $A \in \mathrm{GL}_l(\mathbb{Z})$ is unipotent and $N = A - I$ has a single Jordan block over $\mathbb{Q}$. If $v \in N^{j-1}\mathbb{Q}^l \cap \mathbb{Z}^l$, where $1 \leq j \leq l$, then*

$$|av|_K \leq C_{A,v,S}(1 + |a|^{1/j}) \qquad (a \in \mathbb{Z})$$

for every fixed finite generating set S of K.

Proof. Choose an integral cyclic vector f_1 for N and set $f_i = N^{i-1} f_1$. These vectors form a rational basis and span a finite-index sublattice $F \subset \mathbb{Z}^l$ preserved by A and A^{-1}. The action on this basis is exactly the action in Lemma 2.1. The subgroup $F \rtimes_A \mathbb{Z}$ thus has words for af_i of length $O_l(1 + |a|^{1/i})$ in its indicated generators. Each of these generators has bounded S-length in K, so the same upper estimates hold in K with a different constant.

The vectors $f_j, \dots, f_l$ span $N^{j-1}\mathbb{Q}^l$. Hence there is an integer $D \geq 1$ such that $Dv = \sum_{i=j}^{l} c_i f_i$ with $c_i \in \mathbb{Z}$. Write $a = Da_0 + b$, where $0 \leq b < D$. Then

$$av = \sum_{i=j}^{l} a_0 c_i f_i + bv.$$

Apply the estimates just proved to the terms in the sum, and absorb the finitely many possibilities for bv into the constant. Since $i \geq j$, each term is bounded by a constant times $1 + |a|^{1/j}$. □

3. The integral semidirect product

Fix an integer $m \geq 4$. Let $V = \mathbb{Z}^m$, with basis $e_1, \dots, e_m$, and define endomorphisms J, Y of V by

$$Je_i = e_{i+1}, \qquad Ye_i = ie_{i+1} \quad (1 \leq i < m), \qquad Je_m = Ye_m = 0.$$

Their matrix commutators satisfy

$$[Y, J^j] = YJ^j - J^jY = jJ^{j+1}. \tag{3.1}$$

Put $M = m!$ and consider the additive group of commuting matrices

$$W = M\mathbb{Z}J \oplus M\mathbb{Z}J^2 \oplus \cdots \oplus M\mathbb{Z}J^{m-1}.$$

Conjugation by $\exp Y$ induces an automorphism T of W. Indeed, iterating (3.1) gives

$$T(J^j) = \exp(\operatorname{ad} Y)J^j = J^j(1-J)^{-j}, \qquad T^{-1}(J^j) = J^j(1+J)^{-j}. \tag{3.2}$$

Both expressions are finite integral series, since $J^m = 0$.

Define

$$H = W \rtimes_T \mathbb{Z}, \qquad (P, a)(Q, b) = (P + T^a Q, a + b),$$

and set

$$\rho(P, a) = \exp(P)\exp(aY). \tag{3.3}$$

The conjugation identity defining T shows that ρ is a representation. It takes values in $\mathrm{GL}_m(\mathbb{Z})$. For $P \in W$, both $\exp(P)$ and $\exp(-P)$ are integral: P is divisible by $M = m!$, which clears all factorial denominators in these finite series. Also,

$$\exp(aY)e_i = \sum_{v=0}^{m-i} a^v \binom{i+v-1}{v} e_{i+v} \tag{3.4}$$

is integral for every $a \in \mathbb{Z}$, including negative a.

Our group is

$$G(m) = V \rtimes_\rho H, \qquad (v, h)(v', h') = (v + \rho(h)v', hh'). \tag{3.5}$$

We identify V and H with their natural subgroups of $G(m)$.

Proposition 3.1. *The group $G(m)$ is torsion-free and finitely generated, has Hirsch length $2m$, and is nilpotent of class m. The vector e_m is central.*

Proof. The successive extensions by the free abelian groups V, W, and $\mathbb{Z}$ prove finite generation and torsion-freeness. Additivity of Hirsch length in these extensions gives

$$h(G(m)) = m + (m-1) + 1 = 2m.$$

To see nilpotency, first note that ρ is faithful. If $\exp(P)\exp(aY) = I$, then $\exp(P) = \exp(-aY)$. Taking finite polynomial logarithms gives $P = -aY$. The first lower diagonal of P has a constant entry, whereas that of Y has entries $1, \dots, m-1$. As $m \geq 4$, this forces $a = 0$ and then $P = 0$.

It follows that

$$(v, h) \longmapsto \begin{pmatrix} 1 & 0 \\ v & \rho(h) \end{pmatrix} \tag{3.6}$$

is an embedding into the lower unitriangular integer matrices of size $m+1$. Thus $G(m)$ has class at most m. Let $z = (MJ, 0) \in H$. The iterated commutator with $m-1$ copies of z and one copy of e_1 is the vector

$$(\exp(MJ) - I)^{m-1} e_1 = M^{m-1} e_m \neq 0.$$

It lies in the mth term of the ordinary lower central series, proving that the class is exactly m. Finally, every $\rho(h)$ fixes e_m, so this vector is central in $G(m)$. □

Fix the generating set consisting of $e_1, \dots, e_m$, the elements $(MJ^j, 0) \in H$ for $1 \leq j < m$, and the stable letter $u = (0, 1) \in H$. Until the final step, word lengths refer to this set or to its indicated subsets in the relevant subgroups.

Lemma 3.2. *For integers w, t with $1 \leq w \leq m-2$ and $t \geq 1$,*

$$|(M(tJ^w + J^{w+1}), 0)|_H \leq C_m(1 + t^{1/w}). \tag{3.7}$$

For every $a \in \mathbb{Z}$,

$$|aM^{m-1} e_m|_{G(m)} \leq C_m(1 + |a|^{1/m}). \tag{3.8}$$

Proof. On W, the operator $T - I$ has one rational Jordan block. Indeed, (3.2) gives

$$(T - I)(MJ^j) = jMJ^{j+1} + \text{higher powers of } J.$$

The chain starting at MJ therefore has successive leading terms $M(i-1)!J^i$, for $1 \leq i < m$. In particular, MJ^w lies in the rational image of $(T-I)^{w-1}$. Apply Lemma 2.2 to $H = W \rtimes_T \mathbb{Z}$ and add the fixed vector MJ^{w+1} to obtain (3.7).

For (3.8), use the subgroup generated by V and $z = (MJ, 0) \in H$. Its action on V is $\exp(MJ)$, and the difference $\exp(MJ) - I$ has one rational Jordan block. Its chain beginning at e_1 ends in $M^{m-1}e_m$. Lemma 2.2, with $j = m$, gives the desired words in this subgroup, whose generators have bounded length in $G(m)$. □

4. Short nonconjugate pairs

From now on fix integers and a prime satisfying

$$m \geq 4, \qquad L = \left\lceil \frac{m-1}{2} \right\rceil \leq w \leq m-2, \qquad p > m, \tag{4.1}$$

and put $r = m - w$. The group $G(m)$ and the prime p remain fixed. For each integer $s \geq 1$, define

$$\begin{gathered} t = p^s, \qquad P_s = MJ^w(tI + J), \\ a_s = (0, (P_s, 0)), \qquad c_s = M^{m-1} p^{s-1} e_m, \qquad b_s = a_s c_s. \end{gathered} \tag{4.2}$$

Here c_s is viewed as a central element of $G(m)$.

Proposition 4.1. *The elements a_s and b_s are nonconjugate in $G(m)$ and satisfy*

$$|a_s|, |b_s| \le C_{m,p} p^{s/w} \qquad (s \ge 1). \tag{4.3}$$

Proof. Lemma 3.2 bounds $|a_s|$ by $C_m(1 + p^{s/w})$ and $|c_s|$ by $C_m(1 + p^{(s-1)/m})$. Since $w < m$ and $b_s = a_s c_s$, these bounds imply (4.3) after enlarging the constant.

For nonconjugacy, write

$$R_s = \exp(P_s) - I = P_s B_s, \qquad B_s = \sum_{v=0}^{m-1} \frac{P_s^v}{(v+1)!}. \tag{4.4}$$

Every term of B_s is integral because P_s is divisible by $m!$. Moreover, $B_s - I$ is nilpotent, so its finite geometric inverse is integral. Thus $B_s \in \mathrm{GL}_m(\mathbb{Z})$ and

$$R_s V = P_s V. \tag{4.5}$$

Put $h_s = (P_s, 0) \in H$. If $(v, k) \in G(m)$ conjugates a_s to b_s, equality in the H coordinate requires $k h_s k^{-1} = h_s$. Under this condition the translation coordinate of $(v, k) a_s (v, k)^{-1}$ is $(I - \rho(h_s))v = -R_s v$. Conversely, any vector in $R_s V$ is obtained in this way by a conjugator in V. Therefore

$$a_s \sim_{G(m)} b_s \quad \Longleftrightarrow \quad c_s \in R_s V = P_s V. \tag{4.6}$$

Let $f = J^w(tI + J)$. Its nonzero square block, from the input coordinates $e_1, \dots, e_r$ to the output coordinates $e_{w+1}, \dots, e_m$, is

$$\begin{pmatrix} t & 0 & \cdots & 0 \\ 1 & t & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & 1 & t \end{pmatrix}.$$

Solving $fx = k e_m$ over $\mathbb{Z}$ successively gives $x_1 = \cdots = x_{r-1} = 0$ and $t x_r = k$. Hence $k e_m \in fV$ if and only if t divides k. Since $P_s = Mf$, membership of c_s in $P_s V$ would require

$$p^s \mid M^{m-2} p^{s-1}.$$

This is impossible because $p > m$ and $M = m!$. Equation (4.6) proves the assertion. □

5. Lower bounds for arbitrary finite quotients

Let $\varphi : G(m) \twoheadrightarrow Q$ be any finite quotient in which $\varphi(a_s)$ and $\varphi(b_s)$ are nonconjugate. We shall bound $|Q|$ by examining the finite abelian normal subgroup $A = \varphi(V)$.

5.1. **The obstruction in the p-primary component.** Conjugation by $\varphi(a_s)$ acts on A as $I + R_s$, so its difference from the identity acts as R_s. If $\varphi(c_s)$ belonged to $R_s A$, an element of A would conjugate $\varphi(a_s)$ to $\varphi(b_s)$. Consequently,

$$\varphi(c_s) \notin R_s A. \tag{5.1}$$

On the other hand, there is an exact identity in V:

$$R_s(M^{m-2} e_{m-w}) = M^{m-1} p^s e_m = p c_s. \tag{5.2}$$

To check it, the first application of P_s gives the displayed multiple of e_m, and every further application is zero.

Let A_p be the p-primary component of A, and let $\pi : V \twoheadrightarrow A_p$ be the composition of $\varphi|_V$ with the primary projection. Every endomorphism of A preserves its primary components. On a

component at a prime other than p, multiplication by p is invertible. Equation (5.2) thus puts the corresponding component of $\varphi(c_s)$ in the image of R_s. It follows from (5.1) that

$$\pi(c_s) \notin R_s A_p. \tag{5.3}$$

For the remaining module calculations, use the ring

$$\mathcal{R} = \mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : \ a, b \in \mathbb{Z}, \ p \nmid b\}.$$

Any finite abelian p-group is naturally an $\mathcal{R}$-module, because multiplication by b is invertible whenever $p \nmid b$. The map π extends uniquely to a surjective $\mathcal{R}$-linear map $\mathcal{R}^m \to A_p$. Write Λ for its kernel.

Lemma 5.1. *The submodule Λ is invariant under J and Y. These operators therefore induce endomorphisms of A_p, and*

$$\pi(c_s) \notin f A_p, \qquad f = J^w(tI + J). \tag{5.4}$$

Proof. The matrices $U = \exp(MJ)$ and $E = \exp(Y)$ are the actions of $z = (MJ, 0)$ and $u = (0, 1)$ in $G(m)$. The kernel Λ is therefore invariant under both matrices, and hence under $U - I$ and $E - I$. Since $(U - I)^m = (E - I)^m = 0$, the finite identities

$$\log U = \sum_{v=1}^{m-1} \frac{(-1)^{v+1}}{v} (U - I)^v = MJ, \qquad \log E = \sum_{v=1}^{m-1} \frac{(-1)^{v+1}}{v} (E - I)^v = Y$$

show that Λ is invariant under MJ and Y: every denominator is a unit in $\mathcal{R}$, because $p > m$. As M is also a unit in this ring, Λ is invariant under J.

The matrices B_s and B_s^{-1} in (4.4) are polynomials in J with coefficients in $\mathcal{R}$. They preserve Λ and induce inverse automorphisms of A_p. Multiplication by M is an automorphism as well. Thus $R_s A_p = P_s A_p = f A_p$, and (5.3) gives (5.4). □

5.2. **Matrix units forced by the two actions.** For $1 \le i, j \le m$, let $E_{j,i}$ denote the matrix sending e_i to e_j and all other basis vectors to zero.

Lemma 5.2. *If $j - i \ge L = \lceil (m-1)/2 \rceil$, then $E_{j,i}$ belongs to the associative $\mathcal{R}$-algebra generated by J and Y. In particular, it preserves Λ and induces an endomorphism of A_p.*

Proof. Fix a gap $\ell \ge L$. For $0 \le v \le \ell$, the operator $J^{\ell - v} Y^v$ sends e_i, for $1 \le i \le m - \ell$, to

$$i(i+1) \cdots (i+v-1) e_{i+\ell}.$$

For $v = 0$ the coefficient is 1. The coefficient polynomials are monic of degrees $0, \dots, \ell$, so form an $\mathcal{R}$-basis for the polynomials of degree at most ℓ.

There are $m - \ell$ active input indices, and $m - \ell - 1 \le \ell$. Given one of these indices i_0, the Lagrange polynomial

$$\prod_{\substack{1 \le i \le m-\ell \\ i \ne i_0}} \frac{X - i}{i_0 - i}$$

has degree at most ℓ and takes the value 1 at i_0 and 0 at every other active index. Its denominators are units in $\mathcal{R}$, since all their nonzero factors have absolute value less than $m < p$. Expressing this polynomial in the preceding basis gives an $\mathcal{R}$-linear combination of the operators $J^{\ell - v} Y^v$ that is exactly $E_{i_0 + \ell, i_0}$. Every inactive input index is killed by each of these operators. The final assertion follows from Lemma 5.1. □

5.3. **Independent characters of large order.**

Proposition 5.3. *Under (4.1), every finite quotient Q separating a_s and b_s satisfies*

$$|Q| \geq p^{s(m-w)(w+1-L)}. \tag{5.5}$$

Proof. By (5.4), $\pi(c_s)$ is nonzero in the finite abelian group A_p/fA_p. Characters into $\mathbb{Q}/\mathbb{Z}$ separate points of a finite abelian group: this follows by decomposing the group into cyclic summands and using the standard characters on those summands. We can therefore choose

$$\lambda : A_p \longrightarrow \mathbb{Q}/\mathbb{Z}$$

which vanishes on fA_p and satisfies $\lambda(\pi(c_s)) \neq 0$. Write $\bar{e}_i = \pi(e_i)$, $\lambda_i = \lambda(\bar{e}_i)$, and $\alpha = \lambda_{w+1}$.

The successive nonzero columns of $f = J^w(tI + J)$ give

$$t\lambda_{w+i} + \lambda_{w+i+1} = 0 \quad (1 \leq i < r), \qquad t\lambda_m = 0.$$

Thus

$$\lambda_{w+i} = (-t)^{i-1}\alpha \quad (1 \leq i \leq r). \tag{5.6}$$

Since $c_s = M^{m-1}p^{s-1}e_m$ and M is prime to p, the nonzero value of $\lambda(\pi(c_s))$, together with $p^s\lambda_m = 0$, shows that λ_m has exactly order p^s. Equation (5.6) then implies

$$\operatorname{ord}(\alpha) = p^{sr}. \tag{5.7}$$

Indeed, $\lambda_m = (-p^s)^{r-1}\alpha$, and every value of a character on A_p has order a power of p.

Put $q = w + 1 - L$. For $1 \leq i \leq q$, the gap $w + 1 - i$ is at least L, so Lemma 5.2 makes

$$\chi_i = \lambda \circ E_{w+1,i}$$

a well-defined character of A_p. Its values on the generating elements $\bar{e}_j$ are

$$\chi_i(\bar{e}_j) = \begin{cases} \alpha, & j = i, \\ 0, & j \neq i. \end{cases}$$

In particular, each χ_i has exactly order p^{sr}. If $\sum_{i=1}^{q} b_i\chi_i = 0$, evaluating on $\bar{e}_i$ gives $b_i\alpha = 0$, hence $p^{sr} \mid b_i$ for every i. The characters $\chi_1, \ldots, \chi_q$ therefore generate a subgroup isomorphic to $(\mathbb{Z}/p^{sr}\mathbb{Z})^q$ in the character group of A_p. A finite abelian group and its character group have the same order, so

$$|Q| \geq |A_p| \geq p^{srq}.$$

Since $r = m - w$ and $q = w + 1 - L$, this is (5.5). □

6. Explicit finite separating quotients

The preceding lower bound applies to every finite separating quotient. For completeness, the same matrices give a finite separating quotient for each selected pair directly.

Proposition 6.1. *The image of the affine representation (3.6) modulo p^{sr} is a finite quotient of $G(m)$ in which a_s and b_s remain nonconjugate.*

Proof. On $V/p^{sr}V$, define an additive character μ by

$$\mu(e_i) = 0 \quad (1 \leq i \leq w), \qquad \mu(e_{w+i}) = \frac{(-p^s)^{i-1}}{p^{sr}} + \mathbb{Z} \quad (1 \leq i \leq r).$$

These values are annihilated by p^{sr}, so the character is well-defined. The adjacent columns of f give cancellation, and its last nonzero column gives

$$p^s\mu(e_m) = (-1)^{r-1} + \mathbb{Z} = 0.$$

Hence μ vanishes on fV, and therefore on $R_sV = P_sB_sV = MfB_sV$. In contrast,

$$\mu(c_s) = \frac{(-1)^{r-1}M^{m-1}}{p} + \mathbb{Z} \neq 0. \tag{6.1}$$

Write $E_s = \exp(P_s)$. The matrices of a_s and b_s are respectively

$$\begin{pmatrix} 1 & 0 \\ 0 & E_s \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ c_s & E_s \end{pmatrix}.$$

If their reductions were conjugate in the finite affine image, a conjugating matrix would have the form $\left(\begin{smallmatrix} 1 & 0 \\ v & K \end{smallmatrix}\right)$. Equality of the lower-right blocks would require $KE_sK^{-1} = E_s$ modulo p^{sr}. Equality of the translation columns would then require

$$c_s = (I - E_s)v = -R_sv \pmod{p^{sr}V}.$$

Applying μ contradicts (6.1). Thus this finite matrix image separates the pair. □

7. Proof of the main theorem

Fix $d \geq 2$, take $m = 4d$ and $w = 3d$, and set $\mathcal{G}_d = G(4d)$. These choices satisfy (4.1): here

$$L = 2d, \qquad r = d, \qquad w + 1 - L = d + 1.$$

Proposition 3.1 gives class $4d$ and Hirsch length $8d$. Choose one prime $p > 4d$ and keep it fixed. For every $s \geq 1$, Propositions 4.1, 5.3, and 6.1 give nonconjugate elements with

$$|a_s|, |b_s| \leq Cp^{s/(3d)}, \qquad p^{sd(d+1)} \leq D_{\mathcal{G}_d}(a_s, b_s) < \infty, \tag{7.1}$$

where $C \geq 1$ is independent of s. For any other fixed finite generating set S, enlarge C to preserve the first inequality.

Let $n \geq Cp^{1/(3d)}$, and choose the largest integer $s \geq 1$ such that $Cp^{s/(3d)} \leq n$. Maximality gives $n < Cp^{(s+1)/(3d)}$. Raising this inequality to the power $E_d = 3d^2(d+1)$ and using (7.1), we obtain

$$\mathrm{Conj}_{\mathcal{G}_d,S}(n) \geq p^{sd(d+1)} > C^{-E_d}p^{-d(d+1)}n^{E_d}.$$

This proves the asserted bound for all sufficiently large integers n, with constants allowed to depend on d and S.

Finally, suppose an absolute C_0 satisfied $\mathrm{Conj}_G(n) \preceq n^{C_0h(G)^2}$ for every group in question. Choose d so that $3(d+1)/64 > C_0$. In this one fixed group $\mathcal{G}_d$, the lower exponent $3d^2(d+1)$ is strictly greater than $C_0(8d)^2$. The definition of $\preceq$ would bound $\mathrm{Conj}_{\mathcal{G}_d,S}(n)$ by a constant multiple of $n^{C_0(8d)^2}$, contradicting the lower bound as $n \to \infty$. This completes the proof of Theorem 1.1.

Remark 7.1. The source of the exponent is visible in the bidiagonal conjugacy equation. A character annihilating fA_p satisfies $\lambda_m = (-p^s)^{r-1}\lambda_{w+1}$. Detecting the chosen multiple $p^{s-1}e_m$ forces λ_m to have order p^s, and consequently λ_{w+1} to have order p^{sr}. The second action Y, together with J, supplies the matrix units that reproduce this value in $w + 1 - L$ independent characters. Combining their orders with the word estimate of weight w yields the exponent $w(m - w)(w + 1 - L)$. Theorem 1.1 uses $m = 4d$ and $w = 3d$; it does not assert an exact growth exponent for $\mathcal{G}_d$.

References

[1] Jonas Deré and Lukas Vandeputte, *Uniform Upper Bounds for Conjugacy Separability Growth in Nilpotent Groups*, arXiv:2610.03074v1 [math.GR], 2 October 2026. arxiv.org/abs/2610.03074v1.